

Michelle Liu

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EDUCATION

Brown University

Bachelor of Science, Applied Math – Computer Science, GPA: 4.00/4.00

Providence, RI

Sep 2021 – May 2025

EXPERIENCE

Databricks

Software Engineer

Oct 2025 – Present

Mountain View, CA

- Architected and led the design-to-code frontend refactor for Databricks Assistant's tool-calling system, migrating 60+ *agentic tools* to a new **React/TypeScript** framework and redesigning interaction surfaces for data workflows in **Figma**
- Optimized agent performance and reliability by implementing a TTL-based state management system, enhancing notebook context-passing logic, and maintaining a suite of **Playwright** tests and **Grafana** probes to monitor UI health

Figma

Software Engineer Intern

Jun 2025 – Aug 2025

San Francisco, CA

- Developed custom stroke endcap sizing feature within Figma's Core Experience team, with **C++**, **React**, and **TypeScript**
- Engineered geometry and SceneGraph logic to enable precise, user-controlled cap sizes through vector math
- Partnered with designers to iterate the stroke settings panel design, shaping the endcap controls and overall **UI/UX**

Lightspeed

Data Science Intern

Jan 2024 – May 2025

Providence, RI

- Architected an LDA-based restaurant cohorting model to segment 5,000+ *businesses*, optimizing menu pricing elasticity
- Analyzed transactions using **BigQuery** to uncover sales patterns during holidays, informing restaurant strategy
- Developed sales and tip trend analyses using **Google Cloud Platform** Dataform and **SQL**, with Looker for strategic planning
- Led initiative to optimize table pacing, leveraging statistical models to support just-in-time alerts (**scikit-learn**, **pandas**)

Susquehanna International Group (SIG)

Quantitative Trading Intern

Jun 2024 – Aug 2024

New York, NY

- Quantified the correlation between market attention and stock volatility using time-series analysis and hypothesis testing with **Python** and **riptide**
- Executed real-time trading strategies and risk management in simulated **electronic mock trading** sessions

Brown University

Undergraduate Teaching Assistant

Aug 2022 – Dec 2023

Providence, RI

- Assisted over 300 *students* and designed over 10 *assignments* for **NLP**, statistics, and **machine learning** courses

PROJECTS

Jreamboard | React, Node.js, JavaScript, Express 🐙

- Developed a podcast and audio-based social media web application for the Jream Foundation
- Prototyped frontend using **Figma** and implemented UI using **React**, designing a login and audio posts page
- Built backend architecture with **Express** and **PostgreSQL** to store audio and account information

IntelliGin | Python 🐍 🌐

- Developed a Double DQN (Deep Q-Network) **reinforcement learning** model for Gin Rummy, employing experience replay, self-play and adversarial strategies, achieving performance on par with rule-based adversaries and random agents
- Implemented custom environment in RLCard for Gin Rummy, including engineered feature extraction

North Burial Ground | R, HTML/CSS, Adobe Illustrator 🌐

- Partnered with Providence government to publish historical burial trends at historic cemetery using **ggplot2** and Illustrator

AWARDS

Kleiner Perkins Fellow | Engineering Fellow

- Selected from <3% of global applicants for an elite fellowship pairing top technical talent with high-growth startups

Rohn Truell Prize | Division of Applied Mathematics, Brown University

- Awarded annually to a select few graduating seniors who achieved special distinction in Applied Mathematics

AIME Qualifier | American Invitational Mathematics Examination

- Placed in the top 2.5% of national scorers on the American Mathematics Competitions to qualify for the AIME

SKILLS

Languages: Python (proficient); C, C++, SQL, JavaScript, TypeScript, HTML/CSS (intermediate); R (novice)

Frameworks & Tools: React, Figma, Grafana, Playwright, scikit-learn, pandas, TensorFlow, PyTorch, GCP, AWS